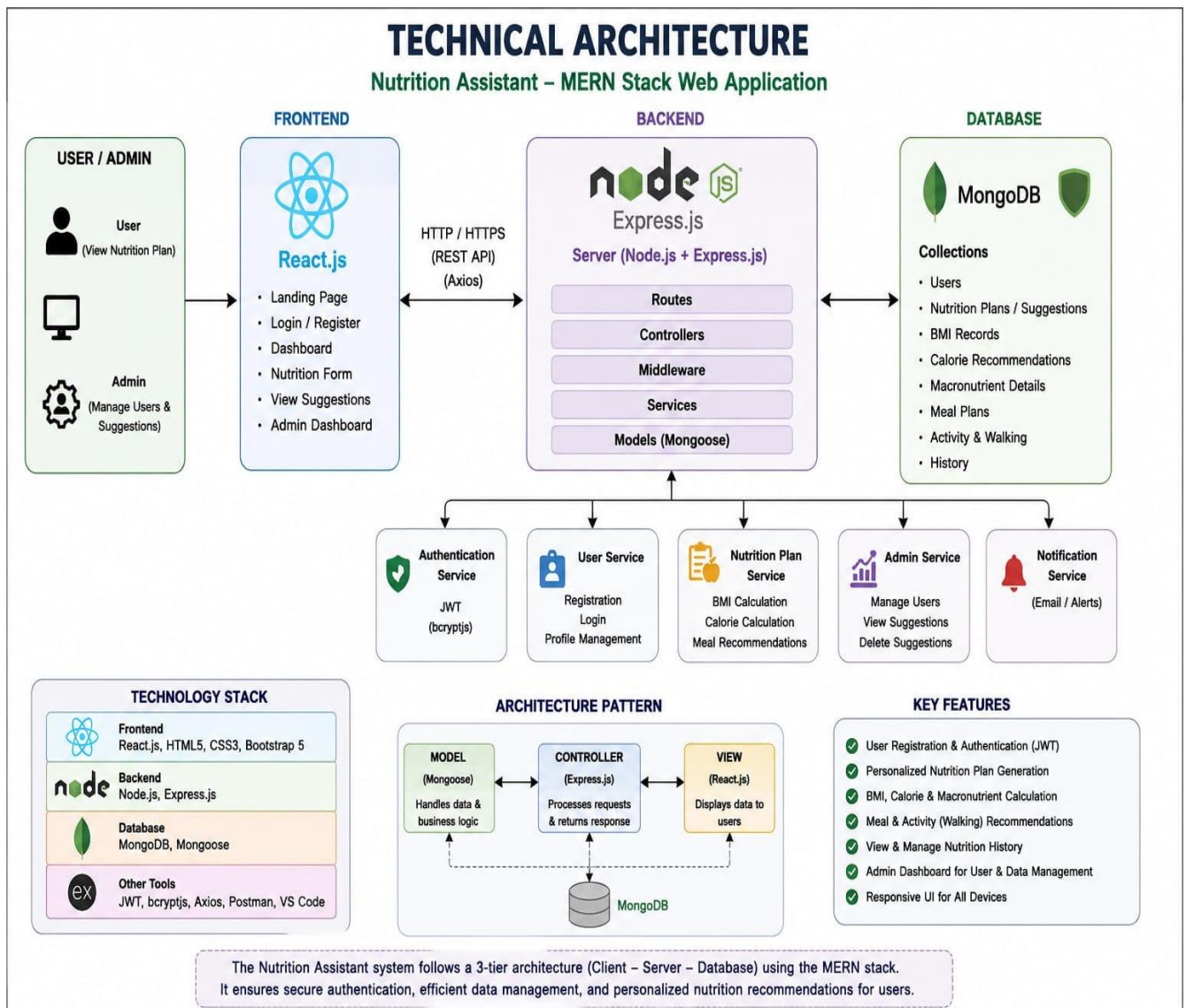


TECHNICAL ARCHITECTURE

The **Nutrition Assistant** is developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js) following the MVC (Model-View-Controller) architecture. The system provides secure user authentication, personalized nutrition recommendations, diet plan management, and an admin dashboard. It ensures efficient communication between the frontend, backend, and database through RESTful APIs.



1. User Interface (UI)

The React.js frontend provides an interactive and responsive interface for users and administrators. Users can register, log in, create personalized nutrition plans, and view diet suggestions, while administrators can manage users and nutrition records.

2. Web Server

The Node.js and Express.js server processes client requests, executes business logic, validates user input, and returns appropriate responses. It acts as the communication bridge between the frontend and MongoDB.

3. API Gateway

RESTful APIs are used for communication between the frontend and backend. APIs handle user authentication, profile management, nutrition suggestion generation, and data retrieval.

4. Authentication Service

The application uses JWT (JSON Web Token) authentication to secure protected routes. Passwords are encrypted using bcryptjs, ensuring user credentials remain secure.

5. Login Service

The login service verifies user credentials, generates a JWT token upon successful authentication, and grants access based on user roles (User or Admin).

6. Registration Service

The registration module collects user information such as name, email, password, age, height, weight, and activity level. Passwords are securely hashed before being stored in MongoDB.

7. Database

MongoDB stores all application data through Mongoose models.

The database includes:

- User details
- Login credentials
- BMI information
- Personalized nutrition suggestions
- Calorie recommendations
- Protein requirements
- Carbohydrate requirements
- Meal timings
- Walking recommendations
- Nutrition history

8. User Profile Management

Users can view and update their personal information, monitor nutrition history, and access previously generated diet plans.

9. Nutrition Plan Generation

Based on user inputs such as age, gender, height, weight, and activity level, the system calculates:

- BMI (Body Mass Index)
- Daily calorie requirement
- Protein requirement
- Carbohydrate requirement
- Meal timing
- Healthy food recommendations
- Walking recommendation
- Personalized nutrition plan

10. View Nutrition Suggestions

Users can view all previously generated nutrition plans, including complete dietary recommendations and health metrics.

11. Admin Dashboard

The administrator can:

- View all registered users
- View all nutrition suggestions
- Delete inappropriate suggestions
- Manage user accounts
- Monitor overall system usage

12. Responsive Design

The application is fully responsive and works seamlessly across desktops, laptops, tablets, and mobile devices using Bootstrap 5, CSS3, and React.js.

This content closely matches the structure and style of your sample while being fully customized for your Nutrition Assistant project.

